

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 11: Geometric Progression

Topic-wise short notes with solved examples

A geometric progression is a sequence in which each term is a fixed multiple of the one before it. This chapter covers its general term, the sum of its terms and the geometric mean.

1 What a G.P. is

- A **geometric progression (G.P.)** is a sequence of non-zero numbers in which the ratio of any term to the term before it is constant.
- The constant is the **common ratio** r , and the first term is a .

$$r = \frac{t_2}{t_1} = \frac{t_3}{t_2} = \dots \quad \text{General G.P.: } a, ar, ar^2, ar^3, \dots$$

- r can be a fraction or a negative number, but it cannot be zero. No term of a G.P. is zero.
- To test for a G.P., check that the ratios of consecutive terms are all equal.

Example 1.1 Which of these are G.P.s? (i) 3, 6, 12, 24, ... (ii) 2, 4, 6, 8, ... (iii) 81, -27, 9, -3, ...

SOLUTION

(i) Ratios: 2, 2, 2. Equal.

(ii) Ratios: $2, \frac{3}{2}, \frac{4}{3}$. Not equal (this is an A.P.).

(iii) Ratios: $-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}$. Equal.

(i) is a G.P. with $r = 2$; (iii) is a G.P. with $r = -\frac{1}{3}$; (ii) is not a G.P.

2 The general term

$$t_n = ar^{n-1}$$

- If the last term is l , the n th term from the end is $\frac{l}{r^{n-1}}$.
- Dividing one term by another removes a : $\frac{t_m}{t_n} = r^{m-n}$.

Example 2.1 Find the 8th term of the G.P. 3, 6, 12, ...

SOLUTION

$$a = 3, r = 2$$

$$t_8 = 3 \times 2^7 = 3 \times 128$$

$$t_8 = 384$$

Example 2.2 Which term of the G.P. 2, 6, 18, ... is 1458?

SOLUTION

$$2 \times 3^{n-1} = 1458, \text{ so } 3^{n-1} = 729 = 3^6$$

$$n - 1 = 6$$

The 7th term.

Example 2.3 The 4th term of a G.P. is 24 and its 7th term is 192. Find the G.P.

SOLUTION

$$ar^3 = 24 \text{ and } ar^6 = 192$$

$$\text{Divide: } r^3 = 8, \text{ so } r = 2 \text{ and } a = \frac{24}{8} = 3$$

The G.P. is 3, 6, 12, 24, ...

3 Sum of n terms

$$S_n = \frac{a(r^n - 1)}{r - 1} \text{ when } r > 1 \quad S_n = \frac{a(1 - r^n)}{1 - r} \text{ when } r < 1$$

- Both forms give the same value; choosing by the size of r keeps the numbers positive.
- When $r = 1$ every term equals a , so $S_n = na$.

Example 3.1 Find the sum of the first 8 terms of the G.P. 1, 2, 4, ...

SOLUTION

$$a = 1, r = 2, n = 8$$

$$S_8 = \frac{1 \times (2^8 - 1)}{2 - 1} = 256 - 1$$

$$S_8 = 255$$

Example 3.2 Find the sum of the first 6 terms of the G.P. 81, 27, 9, ...

SOLUTION

$$a = 81, r = \frac{1}{3}, n = 6$$

$$S_6 = \frac{81 \left(1 - \frac{1}{729}\right)}{1 - \frac{1}{3}} = 81 \times \frac{728}{729} \times \frac{3}{2} = \frac{728}{9} \times \frac{3}{2}$$

$$S_6 = \frac{364}{3} = 121\frac{1}{3}$$

Example 3.3 How many terms of the G.P. 3, 6, 12, ... are needed to give a sum of 381?

SOLUTION

$$\frac{3(2^n - 1)}{2 - 1} = 381, \text{ so } 2^n - 1 = 127 \text{ and } 2^n = 128 = 2^7$$

7 terms.

4 Geometric mean

- If a, G, b are in G.P., then G is the **geometric mean** of a and b .

$$G^2 = ab \quad G = \sqrt{ab}$$

- Three non-zero numbers a, b, c are in G.P. if and only if $b^2 = ac$.

Example 4.1 Find the geometric mean of 4 and 25, and the value of x if $x, 6, 12$ are in G.P.

SOLUTION

$$G = \sqrt{4 \times 25} = \sqrt{100} = 10$$

$$6^2 = 12x, \text{ so } x = 3$$

Geometric mean 10; $x = 3$

5 Choosing terms conveniently

- Three numbers in G.P.: take them as $\frac{a}{r}, a, ar$. Their product is a^3 .

Example 5.1 The product of three numbers in G.P. is 216 and their sum is 26. Find the numbers.

SOLUTION

Let the numbers be $\frac{a}{r}, a, ar$. Product: $a^3 = 216$, so $a = 6$.

Sum: $\frac{6}{r} + 6 + 6r = 26$, so $6r^2 - 20r + 6 = 0$ and $3r^2 - 10r + 3 = 0$

$$(3r - 1)(r - 3) = 0, \text{ so } r = 3 \text{ or } r = \frac{1}{3}$$

The numbers are 2, 6 and 18.

6 A.P. and G.P. compared

	A.P.	G.P.
Constant	Difference d	Ratio r
General term	$a + (n - 1)d$	$a r^{n-1}$
Three terms	$a - d, a, a + d$	$\frac{a}{r}, a, ar$
Condition	$2b = a + c$	$b^2 = ac$
Mean of a and b	$\frac{a + b}{2}$	$\sqrt{ab}$

7 Common mistakes

- Writing $t_n = a r^n$. The power is $n - 1$.
- Finding r as $\frac{t_1}{t_2}$. It is $\frac{t_2}{t_1}$.
- Sign errors with a negative r : odd powers are negative, even powers positive.
- Using the A.P. sum formula for a G.P.